Controlled Diffusions as Unconstrained Smooth Optimization in a Potential: A Hamiltonian-Partials Reformulation with Discrete Dual Pathwise Gradients

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Abstract

We extend the "costate-optimization" reformulation of deterministic optimal control to *stochastic* Itô diffusions. For a d-dimensional state driven by an r-dimensional Brownian motion, we define the extended Hamiltonian $H(x,\lambda,V,\gamma) = \inf_{\alpha\in A}\{\lambda^\top a(x,\alpha) + \frac{1}{2}\operatorname{Tr}[\sigma\sigma^\top(x,\alpha)\,V] + \gamma\, h(x,\alpha)\}$, and reformulate the problem as an *unconstrained* minimization over a deterministic, sufficiently smooth potential $\bar{u} : [0,T]\times\mathbb{R}^d \to \mathbb{R}$. Setting $\lambda = \nabla_x \bar{u}$ and $V = \nabla_x^2 \bar{u}$ yields Markovian (closed-loop) dynamics $\mathrm{d}X_t = \partial_\lambda H(X_t, \nabla\bar{u}, \nabla^2\bar{u}, 1)\,\mathrm{d}t + \Sigma_t\,\mathrm{d}W_t, \quad \Sigma_t\Sigma_t^\top = 2\,\partial_V H(X_t, \nabla\bar{u}, \nabla^2\bar{u}, 1)$, and objective $\mathbb{E}\Big[\int_0^T \partial_\gamma H(X_t, \nabla\bar{u}, \nabla^2\bar{u}, 1)\,\mathrm{d}t + g(X_T)\Big]$. We prove value equivalence with the classical stochastic control problem under standard hypotheses, and derive computable *pathwise* derivatives of the discrete objective using the *discrete dual* of Szepessy–Tempone–Zouraris (CPAM 2001), thus avoiding backward SDEs. Concavity of H in (λ, V, γ) enables PSD-preserving smoothing via piecewise-affine bundles and log-sum-exp, ensuring Lipschitz gradients. We present a practical stochastic-optimization template (SGD/SVRG/mini-batch quasi-Newton) and state-constraint handling through a tangential second-order Hamiltonian with viscosity boundary conditions.

1 Introduction

Solving stochastic optimal control (SOC) problems governed by Itô diffusions is notoriously challenging. On one hand, dynamic programming via the Hamilton–Jacobi–Bellman (HJB) partial differential equation provides a principled approach, but grid-based solutions suffer from the curse of dimensionality in high-dimensional state spaces [4, 8]. On the other hand, indirect methods based on Pontryagin's maximum principle [12, 18] lead to forward–backward simulations (state forward, costate backward) often formulated as coupled forward-backward SDEs [10]. These two-point boundary shooting methods can be numerically unstable and require accurate initialization of the adjoint process. Trajectory optimization and reinforcement learning (RL) techniques offer an alternative by simulating sample paths, but naive approaches can struggle to provide guarantees of optimality and typically yield only open-loop controllers. There is a clear need for frameworks that combine the strengths of classical stochastic control theory with the flexibility of simulation-based learning.

Unlike classical policy gradient methods that differentiate through the simulation or rely on likelihood-ratio estimators, our approach uses the structure of the Hamiltonian to obtain exact, low-variance gradients without backward SDEs.

Inspired by recent advances in deterministic optimal control reformulation [17], we approach SOC from a new angle: we recast the control problem as an *unconstrained smooth optimization* in the space of a suitably defined potential function. In the deterministic (ODE) setting, such a costate-potential reformulation removes the need for solving a two-point boundary value problem by instead optimizing over a value potential whose gradient plays the role of the costate [17]. We extend this philosophy to *controlled Itô diffusions*, introducing an *extended Hamiltonian* $H(x, \lambda, V, \gamma)$ that encapsulates the drift, diffusion, and running cost at state x for a given costate λ, Hessian V, and scalar multiplier γ. Optimizing over a deterministic time-space potential $\bar{u}(t, x)$ (with $\nabla_x \bar{u}$ as costate and $\nabla_x^2 \bar{u}$ as Hessian) yields a Markovian closed-loop control law $\alpha^*(t, x)$ via the minimizer of H, and transforms the SOC problem into a smooth stochastic optimization. In particular, the objective can be written as the expected accumulated cost $\mathbb{E}\Big[\int_0^T \partial_\gamma H\big(X_t, \nabla_x \bar{u}, \nabla_x^2 \bar{u}, 1\big)\, dt + g(X_T)\Big]$, which depends on $\bar{u}$ only through its spatial derivatives. Crucially, because H is defined as an infimum over controls (see (1) below) and is concave in (λ, V, γ), one can apply envelope theorems (Danskin's theorem) to differentiate this objective with respect to $\bar{u}$ without explicitly differentiating through the control or the stochastic dynamics. This leads to a *pathwise gradient* of the SOC objective with respect to the potential, enabling gradient-based optimization of $\bar{u}$.

Building on the above formulation, this paper makes the following core contributions:

1. We develop a *differentiable reformulation* of the SOC problem using the extended Hamiltonian H. This reformulation establishes a value-equivalent optimization in which the search over control policies is replaced by an unconstrained search over a smooth potential function $\bar{u}(t, x)$, whose optimal value coincides with the classical value function $U(t, x)$ under standard regularity conditions.

2. We derive *on-policy* stochastic error identities and a second-order expansion (denoted S-ER and S-ER2) for the value function under a given candidate $\bar{u}$. These analytical expansions quantify the gap between the performance of the current policy (induced by $\bar{u}$) and the optimal policy, providing a foundation for both rigorous performance evaluation and policy improvement. In particular, the S-ER2 expansion leverages second-order information (analogous to a local quadratic expansion of the value function) to guide policy improvement steps.

3. We propose a *discrete dual gradient method* for computing pathwise policy gradients, adapted from the adaptive weak approximation technique of Szepessy–Tempone–Zouraris [16]. By introducing a discrete adjoint (dual) process alongside forward SDE simulation, we obtain unbiased gradient estimates for the parameters of $\bar{u}$ (or the policy) without solving any backward SDE. This yields a practical stochastic gradient (or quasi-Newton) algorithm for policy optimization that operates on simulated trajectories (on-policy data) and is compatible with variance-reduction and mini-batch techniques.

4. We introduce an *adaptive enrichment* strategy for the potential approximator $\bar{u}_\theta$. Using curvature-weighted state visitation diagnostics derived from the S-stochastic error representation (S-ER)/S-ER2 analysis, the method identifies regions of the state space where the current basis is insufficient and augments the function approximator (adding neurons, basis functions, etc.) on-the-fly in regions where the error is high. This curvature-guided refinement concentrates representational capacity where it most improves performance, enabling the learning architecture to increase controller accuracy in a targeted, principled manner.

Our approach bridges several research themes in stochastic control, reinforcement learning, and numerical analysis. In classical stochastic control, two main tools are the HJB equation and the

stochastic maximum principle. The HJB formulation [4, 11] yields a nonlinear second-order PDE for the value function $U(t, x)$; solving it directly is infeasible in large dimensions, but it provides a verification principle and connects to viscosity solutions and dynamic programming. The Pontryagin stochastic maximum principle [18] offers an alternative by characterizing optimal controls via a forward-backward system, introducing an adjoint BSDE for the costate [10]. Our formulation can be viewed as an "open-loop to closed-loop" conversion that avoids both curse-of-dimensionality grids and explicit backward adjoints: by optimizing over the potential $\bar{u}(t, x)$, we ensure the resulting control is feedback (closed-loop) and sidestep continuous-time backward integration. In spirit, this approach shares goals with value-gradient methods in RL [3, 5], which adjust policies by propagating value function gradients through the dynamics. Here, the extended Hamiltonian provides a rigorous foundation for such ideas: we obtain policy gradient formulas that are exact (in the time-discretization limit) and that exploit the structure of H for efficiency.

From a numerical optimization perspective, our on-policy error expansions relate closely to *a posteriori* error estimation techniques. The identities behind S-ER and S-ER2 are analogous to adjoint-based error estimates used to assess solution quality in optimal control solvers [16, 7]. We use these expansions not only to bound the suboptimality of a given $\bar{u}_\theta$, but also to inform the adaptive refinement of $\bar{u}_\theta$ itself–thereby uniting error control with function approximation. Moreover, by incorporating second-order information, our method connects to the family of trajectory-optimization algorithms such as Differential Dynamic Programming (DDP) and iterative LQR, which use local quadratic models of the value function to improve policies [6, 9]. In contrast to classical DDP/iLQR–which require explicit Jacobians and Hessians of the dynamics and cost–our approach obtains analogous curvature information implicitly via the extended Hamiltonian and trajectory simulations.

Finally, our methodology leverages modern function-approximation techniques to handle complex or high-dimensional problems. The potential $\bar{u}(t, x)$ can be represented with flexible architectures (e.g. neural networks or kernel methods), and our curvature-weighted enrichment scheme complements existing basis-adaptation ideas in machine learning. For example, adding basis functions in a reproducing kernel Hilbert space or injecting additional random Fourier features [13] on the fly can be guided by the visitation-weighted error metrics, focusing model capacity on crucial state-space regions. This adaptive, data-driven refinement of the controller's function class is a step toward *certifiable learning-based control*: as the approximation improves, the same error identities provide quantitative certificates of near-optimality (or violation of constraints) for the learned policy.

By formulating stochastic control as a differentiable optimization problem, our work aims to help bridge reinforcement learning and numerical SOC. It enables the use of gradient-based learning algorithms while preserving the theoretical structure (Hamiltonian, costate, etc.) that underpins optimality guarantees. We believe this synergy can facilitate new algorithms that are both data-efficient and equipped with *rigorous diagnostics* for performance–features largely missing in standard RL. In particular, the error estimators we derive allow one to evaluate how far a learned policy is from the optimum and to adaptively improve it, lending greater trustworthiness to learning-based controllers. Moreover, the idea of enriching function approximations in response to error feedback paves the way for *adaptive, certifiable approximation architectures*, wherein controllers (e.g. neural or kernel-based) grow in complexity as needed and come with built-in measures of their improvement. In summary, this work contributes a new theoretical and algorithmic framework for SOC that marries the exploratory power of simulation-based learning with the guarantees of control theory, opening avenues for high-dimensional stochastic control problems that were previously intractable.

2 Problem statements and value equivalence

Notation. Let $\mathbb{R}^d$ be the state space, $\mathbb{S}^d$ the set of $d \times d$ real symmetric matrices, and A a nonempty control set. For matrices $A, B \in \mathbb{S}^d$ write $A : B := \mathrm{Tr}(A^\top B)$. For vectors, $\langle \cdot, \cdot \rangle$ is the Euclidean product and $\| \cdot \|$ the Euclidean norm. The Frobenius norm is $\| \cdot \|_{\mathrm{F}}$. We use $\partial_\lambda H$, $\partial_V H$, $\partial_\gamma H$ for partial derivatives of H w.r.t. its (λ, V, γ) arguments and $D^2_{(\lambda,V)} H$ for the block Hessian in (λ, V). Expectations are with respect to the underlying probability space carrying a d–dimensional Brownian motion $(W_t)_{t\in[0,T]}$. $\mathbb{F}$ its augmented natural filtration, and $\mathcal{P}$ the progressive σ-algebra.

2.1 Problem data and Hamiltonian

Let $a : \mathbb{R}^d \times A \to \mathbb{R}^d$, $\sigma : \mathbb{R}^d \times A \to \mathbb{R}^{d\times r}$ and $h : \mathbb{R}^d \times A \to \mathbb{R}$ be given data with standard linear–growth and local Lipschitz properties in x. The extended Hamiltonian is, for $(x, \lambda, V, \gamma) \in \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{S}^d \times \mathbb{R}$,

$$H(x, \lambda, V, \gamma) \;:=\; \inf_{\alpha\in A} \Big\{ \langle \lambda, a(x, \alpha)\rangle + \tfrac{1}{2}\,\mathrm{Tr}\big[(\sigma\sigma^\top)(x, \alpha)\, V\big] + \gamma\, h(x, \alpha) \Big\}. \tag{1}$$

The value function $U \in C^{1,2}([0, T] \times \mathbb{R}^d)$ solves the HJB equation $U_t + H\big(x, \nabla U, \nabla^2 U, 1\big) = 0$ with terminal condition $U(T, \cdot) = g(\cdot)$.

2.2 Standing assumptions

Assumption 1 (Regularity, computability, envelope)**.** *Fix once and for all either of the following frameworks:*

1. Smooth Hamiltonian. *The Hamiltonian H in* (1) *is C^2 in (λ, V) and C^1 in γ, and for every (x, λ, V, γ) there exists a minimizer $\alpha^*(x, \lambda, V, \gamma) \in A$ attaining the infimum in* (1)*.*

2. Regularized (smoothed) Hamiltonian. *There is a family $(H_\delta)_{\delta>0}$ with $H_\delta \to H$ uniformly on compacts as $\delta \downarrow 0$ such that each H_δ satisfies the properties in item (1). (For instance, PA–bundle/inf–convolution following Sandberg–Szepessy.) All statements below are made for H_δ, with explicit $O(\delta)$ (and bundle) biases recorded where relevant.*

In either case, we assume:

3. ***Exact computability.*** *The minimization in* (1) *(or for H_δ) can be solved exactly to any prescribed accuracy, yielding a measurable selector α^*.*

4. ***Envelope/Danskin identities.*** *For all (x, λ, V, γ) at which H is differentiable in (λ, V, γ) we have*

$$\partial_\lambda H(x, \lambda, V, \gamma) = a\big(x, \alpha^*(x, \lambda, V, \gamma)\big), \quad \partial_V H(x, \lambda, V, \gamma) = \tfrac{1}{2}\,(\sigma\sigma^\top)\big(x, \alpha^*(x, \lambda, V, \gamma)\big),$$

$$\partial_\gamma H(x, \lambda, V, \gamma) = h\big(x, \alpha^*(x, \lambda, V, \gamma)\big).$$

Moreover, we have the Hamiltonian identity

$$H(x, \lambda, V, \gamma) = \langle \lambda, \partial_\lambda H(x, \lambda, V, \gamma)\rangle + \mathrm{Tr}\big[\partial_V H(x, \lambda, V, \gamma)\, V\big] + \gamma\, \partial_\gamma H(x, \lambda, V, \gamma) \tag{2}$$

5. ***Concavity and positivity.*** *H is concave in (λ, V, γ); hence $-D^2_{(\lambda,V)} H \succeq 0$ and $\partial_V H(x, \lambda, V, \gamma) \in \mathbb{S}^d_+$.*

6. ***Closed–loop SDE well–posedness.*** *For any $u \in C^{1,2}([0,T]\times\mathbb{R}^d)$ the on policy SDE*

$$d\bar{X}_t = \partial_\lambda H\big(\bar{X}_t, \nabla u(t,\bar{X}_t), \nabla^2 u(t,\bar{X}_t), 1\big)\,dt + \Sigma_t\,dW_t, \qquad \Sigma_t\Sigma_t^\top = 2\,\partial_V H\big(\bar{X}_t, \nabla u, \nabla^2 u, 1\big), \tag{3}$$

admits a unique strong solution on $[0,T]$.

Remark 1 (One–time disclaimer). Under Assumption 1, all pointwise quantities $\partial_\lambda H$, $\partial_V H$, $\partial_\gamma H$ and the block Hessian $D^2_{(\lambda,V)}H$ are well defined wherever they are used (§**??**–**??** below). If the smoothed H_δ is used, every identity is valid for H_δ, with the explicit smoothing/bundle bias terms stated once in §**??**, and not repeated thereafter.

2.3 Immediate consequences

On-policy cost functional. For any potential $\bar{u} \in C^{1,2}$ and associated on-policy state $\bar{X}$ from (3),

$$J(\bar{u}) \;=\; \mathbb{E}\Big[\int_0^T \partial_\gamma H\big(\bar{X}_t, \nabla\bar{u}(t,\bar{X}_t), \nabla^2\bar{u}(t,\bar{X}_t), 1\big)\,dt + g(\bar{X}_T)\Big].$$

Concavity inequality (supporting plane). For any x and any pairs (λ_0, V_0), (λ_1, V_1),

$$H(x,\lambda_1,V_1,1) \;\le\; H(x,\lambda_0,V_0,1) + \partial_\lambda H(x,\lambda_0,V_0,1)\cdot(\lambda_1-\lambda_0) + \partial_V H(x,\lambda_0,V_0,1):(V_1-V_0).$$

This is used once to deduce the nonnegativity of the on-policy error density in §**??**.

2.4 The classical stochastic control problem (P1-SDE)

Given $T>0$, $a:\mathbb{R}^d\times A\to\mathbb{R}^d$, $\sigma:\mathbb{R}^d\times A\to\mathbb{R}^{d\times r}$, $h:\mathbb{R}^d\times A\to\mathbb{R}$, and $g:\mathbb{R}^d\to\mathbb{R}$, consider

$$\text{(P1-SDE)} \qquad \inf_{\alpha\in\mathcal{A}}\ \mathbb{E}\Big[\int_0^T h(X_t,\alpha_t)\,\mathrm{d}t + g(X_T)\Big] \quad \text{s.t. } (\textbf{??}),\ X_0 = y, \tag{4}$$

where $\mathcal{A}$ denotes progressively measurable, A-valued control processes for which the SDE is well posed and the cost finite.

2.5 Hamiltonian-partials reformulation over a potential (P2-SDE)

We now *restrict* the search to Markovian feedbacks by optimizing over a deterministic potential, namely a value function proxy whose gradients yield the feedback control, $\bar{u}:[0,T]\times\mathbb{R}^d\to\mathbb{R}$ of sufficient smoothness (e.g., $C^{1,2}$). Set

$$\lambda(t,x) := \nabla_x\bar{u}(t,x), \qquad V(t,x) := \nabla_x^2\bar{u}(t,x) \in \mathbb{S}^d. \tag{5}$$

Define the *Hamiltonian-driven* closed-loop SDE

$$\mathrm{d}X_t = \partial_\lambda H\big(X_t, \nabla\bar{u}(t,X_t), \nabla^2\bar{u}(t,X_t), 1\big)\,\mathrm{d}t + \Sigma_t\,\mathrm{d}W_t, \quad \Sigma_t\Sigma_t^\top = 2\,\partial_V H\big(X_t, \nabla\bar{u}(t,X_t), \nabla^2\bar{u}(t,X_t), 1\big), \tag{6}$$

and the objective

$$J(\bar{u}) = \mathbb{E}\Big[\int_0^T \partial_\gamma H\big(X_t, \nabla\bar{u}(t,X_t), \nabla^2\bar{u}(t,X_t), 1\big)\,\mathrm{d}t + g(X_T)\Big]. \tag{7}$$

We pose the stochastic optimization problem

$$\text{(P2-SDE)} \qquad \inf_{\bar{u}\in\mathcal{U}} \; J(\bar{u}) \quad \text{s.t.} \quad (6),\; X_0 = y, \tag{8}$$

where $\mathcal{U}$ is a class of deterministic potentials (e.g., $C^{1,2}$ with polynomial growth) such that the closed-loop SDE is well posed and the cost finite.

Remark 2 (Closed-loop/Markovian control). Let the corresponding optimal control be

$$\alpha^\star(t,x) \in \arg\min_{\alpha\in A}\{\nabla\bar{u}(t,x)^\top a(x,\alpha) + \tfrac{1}{2}\operatorname{Tr}[\sigma\sigma^\top(x,\alpha)\nabla^2\bar{u}(t,x)] + h(x,\alpha)\}$$

which is well-defined by Assumption 1. Then (6) is the state equation under the Markov feedback $\alpha_t = \alpha^\star(t, X_t)$, and the running cost in (7) equals $h(X_t, \alpha^\star(t, X_t))$ by envelope calculus.

2.6 Value equivalence

Let $U_1(0,y)$ be the value of (4). Let $U_2(0,y)$ be the value of (8).

Theorem 1 (Value equivalence and optimal Markov feedback)**.** *Under Assumption 1, $U_2(0,y) = U_1(0,y)$. Moreover, if $U \in C^{1,2}$ is the value function of* (4)*, then the minimizer of* (8) *is $\bar{u} = U$, and the associated feedback $\alpha^\star(t,x)$ from Remark 2 is optimal.*

Proof sketch. We split the proof into two inequalities $U_1(0,y) \le U_2(0,y)$ and $U_2(0,y) \le U_1(0,y)$. For any chosen potential $\bar{u}$, the induced Markov control $\bar{\alpha}(t,x) = \arg\min_\alpha \nabla\bar{u}^\top a + \frac{1}{2}\operatorname{Tr}[\sigma\sigma^\top\nabla^2\bar{u}] + h$ is feasible for the original problem. Using the envelope identities, one sees that $J(\bar{u})$ equals the cost of this Markov policy, hence $U_1 \le J(\bar{u})$ for all $\bar{u}$, implying $U_1(0,y) \le U_2(0,y)$.
Conversely, if $U(t,x)$ is the optimal value function solving the HJB PDE, then U itself is an admissible potential in (P2-SDE). The HJB condition $U_t + H(x, \nabla U, \nabla^2 U, 1) = 0$ guarantees that U attains the infimum in H, so the associated feedback $\alpha^*(t,x)$ realizes the minimum cost. Setting $\bar{u} = U$ gives $J(\bar{u}) = U_1(0,y)$, hence $U_2(0,y) \le U_1(0,y)$. We have used Assumption 1 (measurable selector for the argmin and $U \in C^{1,2}$) to justify applying the HJB verification theorem. By the HJB verification theorem (see, e.g., Fleming & Soner 2006), U attaining the infimum in H implies the feedback control α is optimal. □

Stochastic error representation (on-policy form)

Proposition 1 (Stochastic error representation, on-policy)**.** *Let $U \in C^{1,2}([0,T]\times\mathbb{R}^d)$ solve the HJB equation $U_t + H\big(x, \nabla U, \nabla^2 U, 1\big) = 0$ with terminal condition $U(T,\cdot) = g(\cdot)$, and let $\bar{u} \in C^{1,2}([0,T]\times\mathbb{R}^d)$ be arbitrary. Let $\bar{X}$ solve* (6) *with $\bar{X}_0 = y$. Define the corresponding cost functional*

$$J(\bar{u}) \;=\; \mathbb{E}\Big[\int_0^T \partial_\gamma H\big(\bar{X}_t, \nabla\bar{u}(t,\bar{X}_t), \nabla^2\bar{u}(t,\bar{X}_t), 1\big)\,\mathrm{d}t \;+\; g(\bar{X}_T)\Big]. \tag{9}$$

and let the error integrand be

$$E(t,\bar{x}_t) = \partial_\gamma H(\bar{z}_t) - H\big(\bar{X}_t, \nabla U, \nabla^2 U, 1\big) + \partial_\lambda H(\bar{z}_t)\cdot\nabla U + \partial_V H(\bar{z}_t) : \nabla^2 U.$$

Then the following identity holds:

$$J(\bar{u}) - U(0,y) \;=\; \mathbb{E}\int_0^T \Big[E(t,\bar{x}_t)\Big]\mathrm{d}t, \tag{10}$$

where $\bar{z}_t = (\bar{X}_t, \nabla\bar{u}(t,\bar{X}_t), \nabla^2\bar{u}(t,\bar{X}_t), 1)$ *and* $:$ *denotes* $A : B = \mathrm{Tr}(A^\top B)$.

Moreover, if H *is concave in* (λ, V, γ)*, then the integrand in* (10) *is nonnegative for every* $t \in [0,T]$*, and thus* $J(\bar{u}) \geq U(0,y)$*. Equality holds if and only if* $\nabla\bar{u} = \nabla U$ *and* $\nabla^2\bar{u} = \nabla^2 U$ *along* $\{\bar{X}_t\}_{t\in[0,T]}$.

Proof. Apply Itô's formula to $U(t,\bar{X}_t)$ under the dynamics (6) and recall the HJB equation $U_t + H(x, \nabla U, \nabla^2 U, 1) = 0$. This yields:

$$U(T,\bar{X}_T) - U(0,y) = \int_0^T \Big[-H(\bar{X}_t, \nabla U, \nabla^2 U, 1) + \partial_\lambda H(\bar{z}_t)\cdot\nabla U + \partial_V H(\bar{z}_t) : \nabla^2 U \Big] dt + M_T,$$

where M_T is a martingale with zero expectation. Adding $\int_0^T \partial_\gamma H(\bar{z}_t)dt$ to both sides and taking expectations yields (10).

Observe that $E(t,\bar{x}_t)$ can be rewritten as

$$\begin{aligned} E(t,\bar{x}_t) =& \partial_\gamma H(\bar{x}_t, \nabla\bar{u}(t,\bar{x}_t), \nabla^2\bar{u}(t,\bar{x}_t), 1) - \partial_\gamma H(\bar{x}_t, \nabla U(t,\bar{x}_t), \nabla^2 U(t,\bar{x}_t), 1) \\ &+ \big(\partial_\lambda H(\bar{x}_t, \nabla\bar{u}(t,\bar{x}_t), \nabla^2\bar{u}(t,\bar{x}_t), 1) - \partial_\lambda H(\bar{x}_t, \nabla U(t,\bar{x}_t), \nabla^2 U(t,\bar{x}_t), 1)\big)\cdot\nabla U \\ &+ \big(\partial_V H(\bar{x}_t, \nabla\bar{u}(t,\bar{x}_t), \nabla^2\bar{u}(t,\bar{x}_t), 1) - \partial_V H(\bar{x}_t, \nabla U(t,\bar{x}_t), \nabla^2 U(t,\bar{x}_t), 1)\big) : \nabla^2 U \end{aligned} \tag{11}$$

To see the nonnegativity of the integrand $E(t,\bar{x}_t)$, define the maps:

$$\lambda = \nabla U(t,\bar{X}_t), \qquad \lambda' = \nabla\bar{u}(t,\bar{X}_t),$$
$$V = \nabla^2 U(t,\bar{X}_t), \qquad V' = \nabla^2\bar{u}(t,\bar{X}_t).$$

Then by concavity of H in (λ, V, γ), the function $(\lambda, V, \gamma) \mapsto H(x,\lambda,V,\gamma)$ lies below its supporting hyperplane at $(\lambda', V', 1)$:

$$H(x,\lambda,V,1) \leq H(x,\lambda',V',1) + \partial_\lambda H(x,\lambda',V',1)\cdot(\lambda-\lambda') + \partial_V H(x,\lambda',V',1) : (V-V').$$

Using the Hamiltonian identity (2), this implies:

$$\partial_\gamma H(x,\lambda',V',1) - H(x,\lambda,V,1) + \partial_\lambda H(x,\lambda',V',1)\cdot\lambda + \partial_V H(x,\lambda',V',1) : V \geq 0,$$

which is precisely the integrand $E(t,\bar{x}_t)$ in (10). Equality holds if and only if $\lambda = \lambda'$ and $V = V'$, i.e., $\nabla\bar{u} = \nabla U$ and $\nabla^2\bar{u} = \nabla^2 U$ along the trajectory $\bar{X}_t$.

□

Proposition 2 (Second-order form and curvature weighting)**.** *Assume H is C^2 in (λ, V) (or replace H by its smoothed surrogate H_δ). Define the on-policy errors*

$$\Delta\lambda(t,x) := \nabla U(t,x) - \nabla\bar{u}(t,x), \qquad \Delta V(t,x) := \nabla^2 U(t,x) - \nabla^2\bar{u}(t,x).$$

Then, for each (t,x), a Taylor expansion of $H(x,\cdot,\cdot,1)$ at $(\nabla\bar{u}, \nabla^2\bar{u})$ yields, up to $O((\Delta\lambda, \Delta V)^2)$,

$$\begin{aligned} E(t,\bar{x}_t) \approx \bar{E}(t,\bar{x}_t) &= \partial_\lambda\partial_\gamma H(\tilde{\lambda},\tilde{V}) \cdot \Delta\lambda + \partial_V\partial_\gamma H(\tilde{\lambda},\tilde{V}) : \Delta V \\ &\quad + \Big(\partial^2_{\lambda\lambda} H(\tilde{\lambda},\tilde{V})\,\Delta\lambda + \partial^2_{\lambda V} H(\tilde{\lambda},\tilde{V})\,\Delta V\Big) \cdot \nabla U(t,\bar{x}_t) \\ &\quad + \Big(\partial^2_{V\lambda} H(\tilde{\lambda},\tilde{V})\,\Delta\lambda + \partial^2_{VV} H(\tilde{\lambda},\tilde{V})\,\Delta V\Big) : \nabla^2 U(t,\bar{x}_t) \end{aligned} \tag{12}$$

for some $(\tilde{\lambda},\tilde{V})$ on the segment between $(\nabla\bar{u}, \nabla^2\bar{u})$ and $(\nabla U, \nabla^2 U)$. Consequently,

$$J(\bar{u}) - U(0,y) \;=\; \tfrac{1}{2}\,\mathbb{E}\int_0^T \bar{E}(t,\bar{x}_t)\,\mathrm{d}t \;+\; \text{h.o.t.} \tag{13}$$

Corollary 1 (Second–order structure and two–sided bounds for the error integrand)**.** *Suppose Assumption 1 holds and, in addition, for each $x \in [0,T] \times \mathbb{R}^d$ the map*

$$(\lambda, V) \mapsto H(x,\lambda,V,1)$$

is of class C^2 and concave on $\mathbb{R}^d \times \S^d$. Fix $t \in [0,T]$ and write

$$\lambda^* := \nabla U(t,\bar{X}_t), \quad V^* := \nabla^2 U(t,\bar{X}_t), \quad \bar{\lambda} := \nabla\bar{u}(t,\bar{X}_t), \quad \bar{V} := \nabla^2\bar{u}(t,\bar{X}_t),$$

and

$$\Delta\lambda := \bar{\lambda} - \lambda^*, \qquad \Delta V := \bar{V} - V^*.$$

Then, for each fixed $(t,\bar{X}_t)$, the error integrand in (10) admits the exact representation

$$E(t,\bar{X}_t) = -\int_0^1 (1-s)\begin{pmatrix}\Delta\lambda \\ \Delta V\end{pmatrix}^\top D^2_{(\lambda,V)} H\big(\bar{X}_t, \lambda_s, V_s, 1\big)\begin{pmatrix}\Delta\lambda \\ \Delta V\end{pmatrix}\mathrm{d}s, \tag{14}$$

where

$$\lambda_s := \lambda^* + s\,\Delta\lambda, \qquad V_s := V^* + s\,\Delta V.$$

In particular, by concavity and differentiability of the Hamiltonian, $D^2_{(\lambda,V)} H(x,\cdot,\cdot,1)$ is negative semidefinite and therefore $E(t,\bar{X}_t) \geq 0$.

Moreover, if there exist constants $0 < m \leq M < \infty$ and a neighbourhood $\mathcal{N}$ of the segment

$$\big\{(\lambda_s, V_s) : s \in [0,1]\big\}$$

such that, for all $(x,\lambda,V) \in [0,T] \times \mathbb{R}^d \times \mathcal{N}$ and all $(\eta, W) \in \mathbb{R}^d \times \S^d$,

$$m\big(\|\eta\|^2 + \|W\|_F^2\big) \;\leq\; -\begin{pmatrix}\eta \\ W\end{pmatrix}^\top D^2_{(\lambda,V)} H(x,\lambda,V,1)\begin{pmatrix}\eta \\ W\end{pmatrix} \;\leq\; M\big(\|\eta\|^2 + \|W\|_F^2\big), \tag{15}$$

then we obtain the pointwise two–sided bounds

$$\frac{m}{2}\Big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\Big) \;\leq\; E(t,\bar{X}_t) \;\leq\; \frac{M}{2}\Big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\Big). \tag{16}$$

Proof. Fix $t \in [0,T]$ and $x := \bar{X}_t$. For notational brevity write

$$H(\lambda, V) := H(x, \lambda, V, 1), \quad H_\lambda := \partial_\lambda H, \quad H_V := \partial_V H, \quad H_\gamma := \partial_\gamma H,$$

all evaluated at $(x, \cdot, \cdot, 1)$. Recall that the error integrand in (10) can be written as

$$E(t,x) = H_\gamma(\bar{\lambda}, \bar{V}) + H(\lambda^*, V^*) - H_\lambda(\bar{\lambda}, \bar{V}) \cdot \lambda^* - H_V(\bar{\lambda}, \bar{V}) : V^*. \tag{17}$$

We use the envelope identity (valid for the Hamiltonian defined as the infimum over controls):

$$H(\lambda, V) = H_\lambda(\lambda, V) \cdot \lambda + H_V(\lambda, V) : V + H_\gamma(\lambda, V), \tag{18}$$

which holds at every point where the infimum defining H is attained by some control.

Introduce the straight-line path

$$\lambda_s := \lambda^* + s\,\Delta\lambda, \qquad V_s := V^* + s\,\Delta V, \qquad s \in [0,1],$$

so that $(\lambda_0, V_0) = (\lambda^*, V^*)$ and $(\lambda_1, V_1) = (\bar{\lambda}, \bar{V})$. Define the scalar function

$$F(s) := H(\lambda^*, V^*) - \Big[H(\lambda_s, V_s) + H_\lambda(\lambda_s, V_s) \cdot (\lambda^* - \lambda_s) + H_V(\lambda_s, V_s) : (V^* - V_s)\Big]. \tag{19}$$

Since $H \in C^2$ in (λ, V), F is C^2 on $[0,1]$. At $s = 0$ we have

$$F(0) = H(\lambda^*, V^*) - \big[H(\lambda^*, V^*) + 0 + 0\big] = 0.$$

Next, observe that by concavity of H the function F is nonnegative on $[0,1]$: for each s,

$$H(\lambda^*, V^*) \;\le\; H(\lambda_s, V_s) + H_\lambda(\lambda_s, V_s) \cdot (\lambda^* - \lambda_s) + H_V(\lambda_s, V_s) : (V^* - V_s),$$

so $F(s) \ge 0$. In particular $F(1) \ge 0$.

We now compute $F'(s)$ and $F''(s)$. Differentiate (19) and use the chain rule. First,

$$\frac{d}{ds} H(\lambda_s, V_s) = H_\lambda(\lambda_s, V_s) \cdot \Delta\lambda + H_V(\lambda_s, V_s) : \Delta V.$$

Similarly,

$$\frac{d}{ds}\Big[H_\lambda(\lambda_s, V_s) \cdot (\lambda^* - \lambda_s)\Big] = H_{\lambda\lambda}(\lambda_s, V_s)[\Delta\lambda] \cdot (\lambda^* - \lambda_s) + H_{\lambda V}(\lambda_s, V_s)[\Delta V] \cdot (\lambda^* - \lambda_s) - H_\lambda(\lambda_s, V_s) \cdot \Delta\lambda,$$

and analogously for the H_V–term. A straightforward but somewhat tedious calculation shows that all first–order terms in $(\Delta\lambda, \Delta V)$ cancel, yielding

$$F'(s) = -(1-s) \begin{pmatrix} \Delta\lambda \\ \Delta V \end{pmatrix}^\top D^2_{(\lambda,V)} H(\lambda_s, V_s) \begin{pmatrix} \Delta\lambda \\ \Delta V \end{pmatrix}.$$

(For example, see the derivation of similar identities in dual-weighted residual error analysis.)

Since $F(0) = 0$, integrating $F'(s)$ from 0 to 1 gives

$$F(1) = -\int_0^1 (1-s) \begin{pmatrix} \Delta\lambda \\ \Delta V \end{pmatrix}^\top D^2_{(\lambda,V)} H(\lambda_s, V_s) \begin{pmatrix} \Delta\lambda \\ \Delta V \end{pmatrix} \mathrm{d}s. \tag{20}$$

On the other hand, evaluating (19) at $s = 1$ and using the envelope identity (18) at $(\bar{\lambda}, \bar{V})$, we obtain

$$\begin{aligned} F(1) =& H(\lambda^*, V^*) - \big[H(\bar{\lambda}, \bar{V}) + H_\lambda(\bar{\lambda}, \bar{V}) \cdot (\lambda^* - \bar{\lambda}) + H_V(\bar{\lambda}, \bar{V}) : (V^* - \bar{V})\big] \\ =& H_\gamma(\bar{\lambda}, \bar{V}) + H(\lambda^*, V^*) - H_\lambda(\bar{\lambda}, \bar{V}) \cdot \lambda^* - H_V(\bar{\lambda}, \bar{V}) : V^*, \end{aligned}$$

which is precisely $E(t, x)$ as defined in (17). Combining this with (20) yields the integral representation (14). Because H is concave in (λ, V), $D^2_{(\lambda,V)} H(\lambda_s, V_s)$ is negative semidefinite for all $s \in [0, 1]$, so the integrand in (14) is nonnegative and hence $E(t, \bar{X}_t) \geq 0$.

For the two–sided bounds, assume (15) holds. Then for every $s \in [0, 1]$,

$$m\big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\big) \;\leq\; -\begin{pmatrix} \Delta\lambda \\ \Delta V \end{pmatrix}^\top D^2_{(\lambda,V)} H(\lambda_s, V_s) \begin{pmatrix} \Delta\lambda \\ \Delta V \end{pmatrix} \;\leq\; M\big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\big),$$

and inserting this into (14) gives

$$\begin{aligned} \frac{m}{2}\big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\big) =& m\big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\big) \int_0^1 (1 - s)\,\mathrm{d}s \\ \leq& E(t, \bar{X}_t) \\ \leq& M\big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\big) \int_0^1 (1 - s)\,\mathrm{d}s \\ =& \frac{M}{2}\big(\|\Delta\lambda\|^2 + \|\Delta V\|_F^2\big), \end{aligned}$$

which is exactly (16). This completes the proof. □

Second-order (Taylor) form and curvature weighting

Interpretation. To keep the suboptimality $J(\bar{u}) - U(0, y)$ small, the chosen function class for $\bar{u}_\theta$ must have sufficient expressivity to approximate U in a *path-weighted* H^2 sense–i.e., it must realize small errors in both gradient and Hessian where the policy actually visits. This quantifies the approximation requirement for the parameterization block (polynomial/RKHS/RFF/NN, etc.): richer models (or adaptively enriched bases) are needed whenever the visitation measure $\mu_{\bar{u}}$ concentrates in regions where U has high curvature or fine structure.

Practical surrogate. In code, $\|\bar{u}-U\|^2_{H^2(\mu_{\bar{u}})}$ is estimated by Monte Carlo averages of $\|\nabla(\bar{u}-U)\|^2$ and $\|\nabla^2(\bar{u}-U)\|^2_F$ along simulated on-policy trajectories; this provides a diagnostic for whether the current parameterization has enough capacity, and it guides basis refinement (e.g., add spline knots, kernel centers, or RFF features) in precisely the regions that matter.

Smoothing and bundle surrogates

When H is nonsmooth in (λ, V) (common in bang–bang regimes), replace H by a smoothed PA-bundle model H_δ. Then the derivation above holds verbatim for H_δ, and one adds a bias term of order $\varepsilon_{\mathrm{PA}} + \delta \log m$ coming from the bundle and smoothing errors (see the bundle/smoothing bounds in your ODE draft). Thus

$$J_\delta(\bar{u}) - U(0, y) = \tfrac{1}{2}\,\mathbb{E}\int_0^T \left\langle (\Delta\lambda, \Delta V),\ \left(-D^2_{(\lambda,V)} H_\delta\right)(\Delta\lambda, \Delta V)\right\rangle dt + \mathcal{O}(\varepsilon_{\mathrm{PA}} + \delta \log m) + \text{h.o.t.}, \quad (21)$$

so the same expressivity message applies, with an explicit, controllable modeling bias.

Consequences for the design of the parameterization block

- *What must be approximated?* Not the value U itself, but ∇U and $\nabla^2 U$ in the on-policy L^2 sense. Hence, model families that deliver stable, cheap, and accurate $\nabla\bar{u}_\theta$ and $\nabla^2\bar{u}_\theta$ (and their θ-Jacobians) are most effective (polynomial/spectral, RKHS, RFF, smooth NNs).
- *Where to enrich?* The weighting by the visitation measure $\mu_{\bar{u}}$ implies that basis enrichment should target regions of state space where the current policy spends time (adaptive knots, kernel centers, feature frequencies).
- *How to monitor expressivity?* Track the empirical analogue of $\|\bar{u} - U\|^2_{H^2(\mu_{\bar{u}})}$ using a pilot estimate of U (e.g., from a high-fidelity run, a local HJB fit, or policy-iteration snapshots). In practice, stagnation of the objective together with large variations of $\|\nabla^2\bar{u}_\theta\|$ typically flags under-parameterization.

Deterministic (ODE) case recovered. If $\sigma\sigma^\top \equiv 0$ and V vanishes from H, (**??**) and (**??**) reduce to our previous deterministic representation, including the $\partial^2_{\lambda\lambda}H$-weighted quadratic term.

3 Smoothing and PSD preservation for implementation

Because $H(\lambda, V, \gamma)$ may be non-smooth (e.g. non-differentiable if the infimum in (1) is not unique), we construct an approximating upper envelope of H . Indeed, since H is concave in (λ, V, γ), we can construct a tight piecewise-linear concave approximation of H from its supporting hyperplanes.

Smoothing this bundle via a log-sum-exp or Moreau–Yosida regularization yields a C^2 surrogate H_δ that remains concave and approximates H (within $O(\delta)$ uniformly on compacts). Consequently, for any (t, x) and candidate potential $\bar{u}$, the Hessian term is positive semidefinite: $\partial_V H_\delta(x, \nabla\bar{u}, \nabla^2\bar{u}, 1) \in \S^d_+$, and we can safely set $\Sigma_\delta\Sigma_\delta^\top = 2\partial_V H_\delta =: G_\delta$. This guarantees the simulated diffusion is well-defined (no negative eigenvalues), while also ensuring the gradient of the objective w.r.t. θ is Lipschitz (thanks to the smoothness of H_δ).

This aids convergence of gradient-based solvers.

4 Discrete-time objective and discrete dual pathwise gradients

We discretize (6) by Euler–Maruyama on $t_n = n\Delta t$ with i.i.d. $Z_n \sim \mathcal{N}(0, I_r)$ and write θ for the parameters of a spline/NN representation of $\bar{u}_\theta(t, x)$. Set

$$X_{n+1} = X_n + \tilde{a}_\theta(t_n, X_n)\,\Delta t + \Sigma_\theta(t_n, X_n)\sqrt{\Delta t}\,Z_n, \tag{22}$$

$$J_h(\theta) = \frac{1}{M}\sum_{m=1}^{M}\left[\sum_{n=0}^{N-1}\tilde{h}_\theta\Big(t_n, X_n^{(m)}\Big)\,\Delta t + g\Big(X_N^{(m)}\Big)\right], \tag{23}$$

which empirically approximates $J(\theta) = \mathbb{E}[\dots]$ via M sample trajectories, where

$$\begin{aligned}
\tilde{a}_\theta(t, x) &:= \partial_\lambda H_\delta\big(x, \nabla\bar{u}_\theta(t, x), \nabla^2\bar{u}_\theta(t, x), 1\big),\\
\tilde{Q}_\theta(t, x) &:= 2\,\partial_V H_\delta\big(x, \nabla\bar{u}_\theta(t, x), \nabla^2\bar{u}_\theta(t, x), 1\big), \qquad \Sigma_\theta\Sigma_\theta^\top = \tilde{Q}_\theta,\\
\tilde{h}_\theta(t, x) &:= \partial_\gamma H_\delta\big(x, \nabla\bar{u}_\theta(t, x), \nabla^2\bar{u}_\theta(t, x), 1\big).
\end{aligned}$$

Discrete dual (Szepessy–Tempone–Zouraris). Here $\partial_x\tilde{a}_\theta$ and $\partial_x\Sigma_{\theta,\cdot i}$ denote the Jacobians of the drift and diffusion (with respect to state x), and $\partial_\theta\tilde{a}_\theta$, $\partial_\theta\Sigma_{\theta,\cdot i}$ their sensitivities w.rt. the parameters θ (which involve backpropagating through the network representing $\bar{u}_\theta$). Linearizing (22) gives $\delta X_{n+1} = A_n\,\delta X_n + B_n\,\delta\theta + C_n\,\delta\theta\,Z_n$, with Jacobians

$$\begin{aligned}
A_n &:= I_d + \big(\partial_x\tilde{a}_\theta\big)\Delta t + \sum_{i=1}^{r}\big(\partial_x\Sigma_{\theta,\cdot i}\big)\sqrt{\Delta t}\,Z_{n,i},\\
B_n &:= \big(\partial_\theta\tilde{a}_\theta\big)\Delta t, \qquad C_{n,\cdot i} := \big(\partial_\theta\Sigma_{\theta,\cdot i}\big)\sqrt{\Delta t},
\end{aligned}$$

where all derivatives are evaluated at (t_n, X_n) and depend on $\nabla\bar{u}_\theta$ and $\nabla^2\bar{u}_\theta$ via the chain rule. Define the discrete adjoint $P_n \in \mathbb{R}^d$ backward by

$$\begin{aligned}
P_N &= \nabla g(X_N),\\
P_n &= \mathbb{E}_n\Big[A_n^\top P_{n+1} + \nabla_x\tilde{h}_\theta(t_n, X_n)\,\Delta t\Big], \qquad n = N-1, \dots, 0,
\end{aligned} \tag{24}$$

where $\mathbb{E}_n$ denotes conditional expectation w.r.t. $\sigma\{X_k, Z_k : k \le n\}$; for Monte Carlo we use the pathwise value (replace $\mathbb{E}_n$ by identity), exactly as in [16]. Then the pathwise gradient for (23) is

$$\nabla_\theta J_h = \frac{1}{M}\sum_{m=1}^{M}\sum_{n=0}^{N-1}\left[\big(\partial_\theta\tilde{h}_\theta\big)\Delta t + B_n^\top P_{n+1} + \sum_{i=1}^{r}C_{n,\cdot i}^\top P_{n+1}\,Z_{n,i}\right]^{(m)}. \tag{25}$$

Here the θ-derivatives are obtained by automatic differentiation through $\nabla\bar{u}_\theta$ and $\nabla^2\bar{u}_\theta$; no backward SDE is required. This is a direct specialization of the discrete dual methodology in [16] to our closed-loop SDE with smoothed Hamiltonian.

Remark 3 (On ∂_θ through $\nabla\bar{u}_\theta$ and $\nabla^2\bar{u}_\theta$). If $\bar{u}_\theta$ is a spline or neural network, $\partial_\theta\nabla\bar{u}_\theta$ and $\partial_\theta\nabla^2\bar{u}_\theta$ are available analytically/algorithmically. Since H_δ is C^1-concave, $\partial_\lambda H_\delta$ and $\partial_V H_\delta$ are Lipschitz, so variance of (25) remains controlled under standard growth assumptions.

(As $\nabla_\theta a$ and $\nabla_\theta\Sigma$ are bounded due to the smoothing, the variance of each term in (9) remains controlled, similar in spirit to gradient bounds in stochastic approximation theory [8].)

5 Solver choice: stochastic optimization first

We recommend *stochastic optimization* (e.g., mini-batch SGD/Adam, SVRG/SAGA, or stochastic quasi-Newton) as the default. Deterministic full-batch methods (e.g., L–BFGS) are also applicable when a *sufficiently large* trajectory ensemble is fixed so that sampling noise is negligible. Variance reduction (control variates, common random numbers, antithetics, stratification) and learning-rate schedules are crucial for robustness[8].

In our setting, each gradient evaluation already involves simulating many trajectories; stochastic (incremental) optimization allows continual improvement without waiting for an extremely accurate full-batch gradient, and it naturally handles the noise due to sampling. In contrast, full-batch quasi-Newton methods require a very large fixed sample (to average out noise) and lose the ability to adapt the sample size as learning progresses.

This optimization paradigm is analogous to policy gradient methods in RL, but here the gradient is computed via our pathwise dual method rather than likelihood ratios.

6 State constraints via tangential second-order Hamiltonians

For a closed set $K \subset \mathbb{R}^d$, require $X_t \in K$ a.s. Viability at ∂K demands tangency of drift and diffusion. Let $A_K(x) := \alpha \in A : a(x,\alpha)$ is tangent to K and $\sigma(x,\alpha)$ points into K Define $H_K(x,\lambda,V,1) = \inf_{\alpha\in A_K(x)}\{\lambda^\top a(x,\alpha) + \frac{1}{2}\operatorname{Tr}[\sigma\sigma^\top(x,\alpha)V] + h(x,\alpha)\}$, where $A_K(x)$ enforces the viability constraints. Under appropriate viability conditions, all statements above extend with H replaced by H_K, and dual certification uses the second-order HJB with state-constraint boundary conditions[1, 14, 15, 2].

In other words, $U(t,x)$ solves an HJB equation on K with an inequality constraint $U_t + H_K(x,\nabla U,\nabla^2 U,1) \geq 0$ on the boundary ∂K, ensuring the diffusion reflects or stays tangent; see [Soner 1986].

For example, if $K = [-1,1]^d$, then $A_K(x)$ restricts controls to satisfy $a_i(x,\alpha) \leq 0$ whenever $x_i = 1$ (no outward drift at boundaries).

7 Algorithms

Algorithm 1 On-policy simulator for the closed-loop dynamics induced by $\bar{u}_\theta$

Require: horizon T, steps N, $\Delta t = T/N$; trajectories M; initial law μ_0 (or y deterministic); state domain $D \subset \mathbb{R}^d$; potential $\bar{u}_\theta$; extended Hamiltonian H (with exact inner minimization in α); small $\varepsilon_{\text{psd}} > 0$.

Ensure: Paths $\{X_n^{(m)}\}_{n=0}^N$ and running costs $\{c_n^{(m)}\}$ for $m = 1, \ldots, M$.

1: **for** $m = 1$ to M **do**
2: Sample $X_0^{(m)} \sim \mu_0$ ▷ or set $X_0^{(m)} = y$
3: **for** $n = 0$ to $N - 1$ **do**
4: $x \leftarrow X_n^{(m)}$; $t \leftarrow n\Delta t$
5: Compute $\lambda := \nabla \bar{u}_\theta(t, x)$ and $V := \nabla^2 \bar{u}_\theta(t, x)$ ▷ V forced symmetric in implementation
6: Compute drift $a := \partial_\lambda H(x, \lambda, V, 1)$
7: Compute diffusion metric $G := 2\, \partial_V H(x, \lambda, V, 1)$ ▷ $G \in \mathbb{R}^{d \times d}$; symmetric PSD in theory
8: Stabilize: $G \leftarrow [\text{sym}(G)]_+ + \varepsilon_{\text{psd}} \, \text{Id}$
9: Factor $G = \Sigma \Sigma^\top$ ▷ e.g., eigen- or Cholesky factorization
10: Draw $\xi \sim \mathcal{N}(0, \text{Id}_d)$ and set $\Delta W := \sqrt{\Delta t}\, \xi$
11: $x^{\text{prop}} \leftarrow x + a\, \Delta t + \Sigma\, \Delta W$
12: **State constraints:**
If projection: $X_{n+1}^{(m)} \leftarrow \text{proj}_D(x^{\text{prop}})$
If reflection: reflect x^{prop} at ∂D to get $X_{n+1}^{(m)}$
13: Compute running cost $c_n^{(m)} := \partial_\gamma H(X_n^{(m)}, \lambda, V, 1)\, \Delta t$
14: Save terminal payoff $g^{(m)} := g(X_N^{(m)})$

Remark 4 (Implementation). If the inner minimization in H returns a minimizer $\alpha^\star(t, x)$, it is not needed explicitly here; only $\partial_\lambda H, \partial_V H, \partial_\gamma H$ evaluated at $(x, \nabla \bar{u}, \nabla^2 \bar{u}, 1)$ are used.

In theory G is PSD by construction; in practice we enforce this by symmetrization and a small diagonal lift ε_{psd}.

Step 1 uses an initial law μ_0 or a fixed y. This initial law μ_0 can be a delta (for a fixed start) or an empirical distribution of initial states if evaluating multiple start points.

Algorithm 2 Monte Carlo estimation of $J(\bar{u}_\theta)$ and $\nabla_\theta J(\bar{u}_\theta)$ via a discrete dual

Require: Simulator from Alg. 1; time grid $\{t_n\}$; stored at each step: X_n, a_n, Σ_n and their x- and θ-sensitivities if needed; running cost $c_n(x,\theta) := \partial_\gamma H(x, \nabla\bar{u}_\theta, \nabla^2\bar{u}_\theta, 1)$; terminal cost $g(x)$.

Ensure: $\widehat{J}(\theta)$ and an unbiased pathwise estimator $\widehat{\nabla_\theta J}(\theta)$.

for $m = 1$ to M **do** ▷ *independent trajectories*
  Forward pass: generate $\{X_n^{(m)}\}_{n=0}^N$ and $\{\xi_n^{(m)}\}_{n=0}^{N-1}$ with Alg. 1
  Compute $\widehat{J}^{(m)} \leftarrow \sum_{n=0}^{N-1} c_n(X_n^{(m)}, \theta)\,\Delta t + g(X_N^{(m)})$
  Backward dual (discrete adjoint):
  Initialize $\lambda_N \leftarrow \nabla_x g(X_N^{(m)})$
  for $n = N-1$ down to 0 **do**
    $x \leftarrow X_n^{(m)}$, $\xi \leftarrow \xi_n^{(m)}$, $t \leftarrow n\Delta t$
    Evaluate $\lambda := \nabla\bar{u}_\theta(t,x)$, $V := \nabla^2\bar{u}_\theta(t,x)$
    $a := \partial_\lambda H(x,\lambda,V,1)$, $G := 2\,\partial_V H(x,\lambda,V,1)$, $\Sigma\Sigma^\mathsf{T} = G$
    Compute Jacobians (w.r.t. state):
    $A_n := \mathrm{Id} + \Delta t\,(\partial_x a)(t,x) + \sqrt{\Delta t}\,\sum_{k=1}^r \xi_k\,\partial_x \Sigma_{\cdot k}(t,x)$
    Compute instantaneous gradient of running cost:
    $r_n := \Delta t\,\nabla_x c_n(x,\theta)$
    Update adjoint:
    $\lambda_n \leftarrow A_n^\mathsf{T}\lambda_{n+1} + r_n$
  Gradient accumulation (reverse mode w.r.t. θ):
  Initialize $g_\theta^{(m)} \leftarrow 0$
  for $n = 0$ to $N-1$ **do**
    $B_n := \Delta t\,\partial_\theta a(t_n, X_n^{(m)}) + \sqrt{\Delta t}\,\sum_{k=1}^r \xi_{n,k}^{(m)}\,\partial_\theta \Sigma_{\cdot k}(t_n, X_n^{(m)})$
    $g_\theta^{(m)} \leftarrow g_\theta^{(m)} + \Delta t\,\partial_\theta c_n(X_n^{(m)},\theta) + B_n^\mathsf{T}\lambda_{n+1}$

Outputs: $\widehat{J}(\theta) := \frac{1}{M}\sum_{m=1}^M \widehat{J}^{(m)}$, $\quad \widehat{\nabla_\theta J}(\theta) := \frac{1}{M}\sum_{m=1}^M g_\theta^{(m)}$.

Remark 5. (i) This is the standard discrete adjoint for explicit schemes, avoiding BSDEs; it follows the discrete dual philosophy of *Szepessy–Tempone–Zouraris 2001 (CPAM)* for pathwise/weak?error representation in SDE discretizations (cite this entry in your refs.bib). (ii) Differentiation through Σ uses a stable factorization (eigen or Cholesky). If needed, implement a PSD–safe Jacobian by differentiating the eigen decomposition with eigenvalue clipping.

The above yields an unbiased estimator of the (discrete) gradient at cost $O(N)$ per trajectory (for the backward pass) in addition to the $O(N)$ forward simulation.

(For long time horizons, one may store checkpoints and recompute segments to save memory, akin to backpropagation through time.)

Note that $\partial_\theta c_n$ is computed by propagating the adjoint P_{n+1} through the neural network representing $\bar{u}_\theta$ (one additional reverse pass). This is all handled seamlessly by modern autodiff libraries.

The work [STZ2001] derived a similar adjoint for analyzing weak error of time-stepping schemes; here we repurpose it to obtain policy gradients.

Algorithm 3 PA-bundle construction of H_δ with log-sum-exp smoothing

Require: Set of expansion points $\mathcal{Z} = \{(x^i, \lambda^i, V^i)\}_{i=1}^{N_b}$; smoothing $\delta > 0$; regularization $\eta \geq 0$.
Ensure: Smoothed surrogate $H_\delta(x, \lambda, V, 1)$ and its derivatives in (λ, V).

1: **for** each bundle point $i = 1, \ldots, N_b$ **do**
2: Solve exactly $\alpha^i \in \arg\min_{\alpha \in A} \left\{ \lambda^{i\mathsf{T}} a(x^i, \alpha) + \frac{1}{2} \operatorname{Tr}[(\sigma\sigma^\mathsf{T})(x^i, \alpha) V^i] + h(x^i, \alpha) \right\}$
3: Form affine minorant

$$\ell_i(x, \lambda, V) := \lambda^\mathsf{T} a(x, \alpha^i) + \tfrac{1}{2} \operatorname{Tr}[(\sigma\sigma^\mathsf{T})(x, \alpha^i) V] + h(x, \alpha^i)\,.$$

4: Define PA?envelope $H_{\mathrm{PA}}(x, \lambda, V, 1) := \min_i \ell_i(x, \lambda, V)$
5: Smooth via log?sum?exp:

$$H_\delta(x, \lambda, V, 1) = -\delta \log\Big(\sum_{i=1}^{N_b} \exp\{-\ell_i(x, \lambda, V)/\delta\} \Big) + \tfrac{\eta}{2} \|(\lambda, V)\|^2\,.$$

6: Derivatives:
Let $w_i \propto \exp\{-\ell_i/\delta\}$ and $\sum_i w_i = 1$;
$\partial_\lambda H_\delta = \sum_i w_i\, a(x, \alpha^i) + \eta\, \lambda$;
$\partial_V H_\delta = \frac{1}{2} \sum_i w_i\, (\sigma\sigma^\mathsf{T})(x, \alpha^i) + \eta\, V$;
$\partial_\gamma H_\delta$ is the weighted average of $h(x, \alpha^i)$ (here $\gamma = 1$ in our use).
7: (Optional) **Bundle growth:** add new (x, λ, V) from current policy rollouts and repeat.

Algorithm 4 Stochastic optimization of θ for the SOC problem (proper stochastic optimization)

Require: Initial θ_0; stepsizes $\{\eta_k\}$; batch sizes M_k; simulator (Alg. 1); gradient estimator (Alg. 2); optional bundle surrogate (Alg. 3); constraint handler (Alg. 6).

1: **for** $k = 0, 1, 2, \ldots$ **do**
2: (Optional) update surrogate H_δ using fresh rollout statistics (Alg. 3)
3: Draw mini–batch of M_k trajectories under $\bar{u}_{\theta_k}$
4: Compute $\widehat{J}(\theta_k)$ and $\widehat{\nabla_\theta J}(\theta_k)$
5: $\theta_{k+1} \leftarrow \theta_k - \eta_k\, \widehat{\nabla_\theta J}(\theta_k)$ ▷ SGD/Adam/variance?reduced variants
6: (Optional) adaptively enrich the model for $\bar{u}_\theta$ (Alg. 5) if diagnostics flag under–parameterization
7: **Stop** if relative change of $\widehat{J}$ is small and constraint violation below tolerance

Remark 6. We use proper stochastic optimization. Deterministic solvers may be used only if one fixes a sufficiently large ensemble so that the noise in the gradient is negligible.

Algorithm 5 Visitation-weighted enrichment of the parameterization block for $\bar{u}_\theta$

Require: Current θ; visitation samples $\{(t_n, X_n)\}$ from on-policy rollouts; diagnostic scores s_n (e.g., curvature-weighted residuals or large local variance of gradient estimates); target budget K_{new}.

Ensure: Updated dictionary/parameterization for $\bar{u}_\theta$.

Rank samples by s_n and select top K_{new} points $\{(t_j^\star, x_j^\star)\}$
if polynomial/spectral subspace **then**
    Increase degree locally or add localized polynomials centered at $x_j^\star$
else if RKHS kernel model **then**
    Add new kernel centers at $x_j^\star$ with bandwidth chosen by local curvature/scale
else if Random Fourier Features (RFF) **then**
    Fit/refresh frequency distribution using selected $\{x_j^\star\}$ (e.g., via adaptive resampling)
    Sample K_{new} frequencies and augment the RFF layer
else if Neural parametrization **then**
    Add neurons or widen layers; optionally initialize from local least?squares fit around $x_j^\star$
Re?initialize/continue training from previous θ (warm start)

Algorithm 6 State–constraint enforcement for domain D

Require: Discretized proposal x^{prop}; current x; domain D; option flag.

if Projection **then**
    $x^{\text{new}} \leftarrow \text{proj}_D(x^{\text{prop}})$
else if Reflection **then**
    If $x^{\text{prop}} \in D$ then $x^{\text{new}} \leftarrow x^{\text{prop}}$, else reflect the increment at ∂D
else if Barrier/Penalty **then**
    $x^{\text{new}} \leftarrow x^{\text{prop}}$ and add barrier term to running cost:
    $c \leftarrow c + \rho\, \phi_D(x^{\text{new}})$ with ϕ_D a smooth barrier and $\rho\uparrow$
return x^{new}

Remark 7. An alternative way to handle state constraints is to let the admissible control set depend on $(x, t, \Delta t)$, and choosing $\Delta t(t, x)$ sufficiently small so that the probability of leaving the domain is tiny.

Algorithm 7 Curvature-weighted error diagnostic from the stochastic error representation

Require: On-policy samples (t_n, X_n) under $\bar{u}_\theta$; local derivatives $\nabla\bar{u}_\theta, \nabla^2\bar{u}_\theta$; surrogate of $(-D^2_{(\lambda,V)}H)$ or its smoothed H_δ Hessian.

Ensure: Scalar diagnostic $\widehat{\mathcal{E}}$ and per-sample weights for enrichment.

for each sample (t_n, X_n) **do**
    Estimate $\Delta\lambda_n, \Delta V_n$ using a local proxy of $\nabla U, \nabla^2 U$
    (e.g., from a higher?fidelity snapshot, a local HJB fit, or a previous policy iterate)
    Compute curvature matrix $C_n := -D^2_{(\lambda,V)}H(X_n, \tilde{\lambda}_n, \tilde{V}_n, 1)$
    Score $s_n := \big\langle (\Delta\lambda_n, \Delta V_n),\, C_n\,(\Delta\lambda_n, \Delta V_n)\big\rangle$
$\widehat{\mathcal{E}} \leftarrow \frac{1}{N}\sum_n \frac{1}{2}\, s_n\, \Delta t$ ▷ cf. (S-ER2) ignoring $o(1)$ terms
return $\widehat{\mathcal{E}}$ and $\{s_n\}$ for Algorithm 5

This operationalizes the "S–ER2"representation: curvature?weighted squared errors of $(\nabla\bar{u}_\theta, \nabla^2\bar{u}_\theta)$

along the on?policy visitation measure.

Table 1: Computational building blocks and typical per-iteration costs. Here d is the state dimension, M trajectories, N steps, and C_H denotes the (assumed exact) cost of the inner minimization for H.

Block	What it computes	Cost (typical)
Alg. 1	On-policy rollouts and costs	$O(MN\,(C_H + d^3))$ (PSD factor & derivatives)
Alg. 2	$\widehat{J}$ and $\widehat{\nabla_\theta J}$	$O(MN \cdot \mathrm{Jac}(\bar{u}_\theta))$
Alg. 3	H_δ surrogate and ∂H_δ	$O(N_b\, C_H)$ per eval; build cost $O(N_b)$ solves
Alg. 4	Policy parameter updates	dominated by gradient step above
Alg. 5	Adaptive basis/feature growth	data-driven; typically sublinear in MN
Alg. 6	Constraint handling	negligible vs. rollout; barrier increases cost in c_n
Alg. 7	S-ER2 diagnostic	$O(MN\,d^3)$ (curvature) or cheaper with surrogates

Implementation pipeline. At each outer iteration k, we (i) update the smoothed surrogate H_δ (Alg. 3), (ii) generate on-policy rollouts under $\bar{u}_{\theta_k}$ (Alg. 1), (iii) estimate $J(\bar{u}_{\theta_k})$ and its pathwise gradient via a discrete adjoint [16] (Alg. 2), (iv) take a stochastic optimization step on θ (Alg. 4), (v) enforce constraints along the path (Alg. 6), and (vi) compute a curvature-weighted diagnostic to decide whether to enrich the parameterization (Alg. 7 and Alg. 5).

8 Numerical template and experiments

Algorithm.

1. Choose a parametric potential $\bar{u}_\theta(t, x)$ (tensor-product B-splines or a compactly supported NN) and initialize θ.
2. Build a PA-bundle for H, smooth to H_δ (log-sum-exp). Define $\tilde{a}_\theta, \tilde{Q}_\theta, \tilde{h}_\theta$ from H_δ.
3. Forward simulate (22) with common random numbers and compute $J_h(\theta)$. We suggest using common random numbers to reduce variance between policy iterations during evaluation
4. Compute $\nabla_\theta J_h$ from (24)–(25) (discrete dual); update θ by a stochastic optimizer (e.g. SGD/SVRG). Periodically, anneal $\delta \downarrow 0$ and enrich the PA-bundle. (see Algorithm in Section 8 for bundle update strategy)
5. (Optional) Train an HJB subsolution for a dual lower bound and report a certified gap.

Experiments. We plan several studies: (i) stochastic LQR with box constraints $K = [-1, 1]^2$; (ii) a nonlinear oscillator with control bounds; (iii) chance-constrained diffusion with viability via H_K. We will use stochastic optimizers (SGD/SVRG) and, for comparison, full-batch L–BFGS with large fixed ensembles to emulate deterministic optimization.

8.1 Verification Example: Stochastic LQ Control

To validate the theoretical framework and numerical implementation, we consider the classical linear-quadratic (LQ) stochastic control problem in $\mathbb{R}^2$. The system dynamics are given by

$$dX_t = AX_t\,dt + Bu_t\,dt + \sigma\,dW_t, \qquad X_0 = x \in \mathbb{R}^2, \tag{26}$$

with cost functional

$$J(u) = \mathbb{E}\left[\int_0^T \left(X_t^\top Q X_t + u_t^\top R u_t\right) dt + X_T^\top S X_T\right]. \tag{27}$$

Here, we choose:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \sigma = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix},$$
$$Q = I_2, \quad R = 1, \quad S = I_2, \quad T = 1.$$

The value function has the form $U(t,x) = x^\top P(t) x$, where $P(t)$ solves the Riccati equation

$$-\dot{P} = A^\top P + PA - PBR^{-1}B^\top P + Q, \quad P(T) = S. \tag{28}$$

The optimal feedback control is given by

$$u^*(t,x) = -R^{-1}B^\top P(t) x. \tag{29}$$

The extended Hamiltonian becomes

$$H(x,\lambda,V,\gamma) = \min_{u\in\mathbb{R}} \left\{\lambda^\top (Ax + Bu) + \frac{1}{2}\mathrm{Tr}(\sigma\sigma^\top V) + \gamma\left(x^\top Q x + u^\top R u\right)\right\}, \tag{30}$$

with minimizer

$$u^*(x,\lambda,\gamma) = -\frac{1}{2\gamma} R^{-1} B^\top \lambda. \tag{31}$$

This example admits a fully explicit solution and will be used to verify:

- Correct implementation of the extended Hamiltonian and its partials,
- On-policy error identities (S-ER, S-ER2),
- Pathwise gradient computation via discrete duals,
- Accuracy and convergence of the approximated potential $\bar{u}$ against the known solution U.

8.2 Stochastic LQ Example: Explicit Formulae and Discrete Gradient

We revisit the linear-quadratic stochastic control problem described above, and derive explicit expressions for all ingredients of the Hamiltonian-based formulation.

System and cost:

$$dX_t = AX_t\,dt + Bu_t\,dt + \sigma\,dW_t, \qquad X_0 = x \in \mathbb{R}^2,$$
$$J(u) = \mathbb{E}\left[\int_0^T \left(X_t^\top Q X_t + u_t^\top R u_t\right) dt + X_T^\top S X_T\right],$$

with

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \sigma = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}, \quad Q = S = I_2, \quad R = 1.$$

Hamiltonian and optimal control: Let $\lambda \in \mathbb{R}^2$, $V \in \mathbb{R}^{2\times 2}$, and $\gamma = 1$. Then

$$H(x, \lambda, V, 1) = \min_{u\in\mathbb{R}} \left\{ \lambda^\top (Ax + Bu) + \tfrac{1}{2}\mathrm{Tr}(\sigma\sigma^\top V) + x^\top Qx + u^\top Ru \right\}.$$

The minimizer is

$$u^*(x, \lambda) = -\tfrac{1}{2}R^{-1}B^\top \lambda = -\tfrac{1}{2}B^\top \lambda.$$

Substituting this into the Hamiltonian gives the minimized form

$$H(x, \lambda, V, 1) = \lambda^\top Ax - \tfrac{1}{4}\lambda^\top BR^{-1}B^\top \lambda + \tfrac{1}{2}\mathrm{Tr}(\sigma\sigma^\top V) + x^\top Qx.$$

First-order derivatives: Let $\bar{u}(t, x) = x^\top P(t)x$ be the approximate potential, with $\lambda = \nabla\bar{u} = 2Px$, and $V = \nabla^2\bar{u} = 2P$. Then:

$$\begin{aligned}
\partial_\lambda H &= Ax - \tfrac{1}{2}BB^\top \lambda = Ax - \tfrac{1}{2}BB^\top (2Px),\\
\partial_V H &= \tfrac{1}{2}\sigma\sigma^\top,\\
\partial_\gamma H &= x^\top Qx + (u^*)^2 = x^\top Qx + \tfrac{1}{4}\lambda^\top BB^\top \lambda.
\end{aligned}$$

Euler discretization (forward simulation): Let $\Delta t > 0$, and define:

$$X_{n+1} = X_n + (AX_n + Bu_n)\,\Delta t + \sigma\sqrt{\Delta t}\, Z_n,$$

where $u_n = -\frac{1}{2}B^\top \lambda_n = -B^\top PX_n$, and $Z_n \sim \mathcal{N}(0, I_2)$.

Dual (costate) recursion: Let $P_N = \nabla g(X_N) = 2SX_N$, and define:

$$P_n = \left(I + A^\top \Delta t - 2\Delta t\, PBB^\top\right) P_{n+1} + 2QX_n\Delta t.$$

Pathwise gradient of cost: The stochastic gradient of $J(\bar{u})$ with respect to parameters θ in $P(\theta)$ (e.g. basis function weights) is:

$$\nabla_\theta J = \frac{1}{M}\sum_{m=1}^{M}\sum_{n=0}^{N-1}\left[(\partial_\theta\nabla\bar{u})(t_n, X_n)^T B\, P_{n+1}\Delta t + (\partial_\theta\nabla^2\bar{u})(t_n, X_n) : \sigma\sigma^\top\, \Delta t + \partial_\theta\bar{u}(t_n, X_n)\,\Delta t\right]^{(m)},$$

where all derivatives can be evaluated analytically or via automatic differentiation depending on the parameterization.

This example provides a fully explicit benchmark to validate all components of the formulation: closed-loop simulation, Hamiltonian partials, S-ER/S-ER2 identities, discrete duals, and pathwise gradients.

8.3 High-dimensional basket example with on/off control and terminal payoff

We consider a high-dimensional example closely related to optimal stopping: a basket of d assets evolving under a Black–Scholes diffusion, where a binary on/off control $\alpha_t \in \{0, 1\}$ governs whether the diffusion continues ($\alpha_t = 1$) or the state is frozen ($\alpha_t = 0$). The payoff is realized only at the terminal time T via a basket put payoff $g(X_T)$; there is no running cost.

Controlled SDE and terminal payoff. Let $S_t = (S_t^1, \dots, S_t^d)^\top \in \mathbb{R}_+^d$ denote the (discounted or risk–neutral) asset prices. In the uncontrolled case, they follow a Black–Scholes diffusion

$$\mathrm{d}S_t^i = S_t^i \sum_{k=1}^{d} \sigma_{ik}\, \mathrm{d}W_t^k, \qquad i = 1, \dots, d,$$

where W is a d–dimensional Brownian motion and $\sigma \in \mathbb{R}^{d\times d}$ encodes the instantaneous covariance $(\sigma\sigma^\top)$.

We model the controlled state $X_t \in \mathbb{R}_+^d$ by

$$\mathrm{d}X_t = \alpha_t\big(a(X_t)\,\mathrm{d}t + b(X_t)\,\mathrm{d}W_t\big), \qquad X_0 = x \in \mathbb{R}_+^d, \tag{32}$$

with binary control

$$\alpha_t \in A := \{0, 1\}, \qquad \alpha_t = 1 \text{ ('continue')}, \quad \alpha_t = 0 \text{ ('freeze')}.$$

For concreteness we take

$$a(x) := 0_d, \qquad b(x) := \mathrm{diag}(x)\,\sigma, \qquad x \in \mathbb{R}_+^d,$$

so that X_t coincides with an uncontrolled lognormal basket while $\alpha_t = 1$, and remains constant once α_t switches to 0.

The terminal payoff is a basket put

$$g(x) := \Big(K - \sum_{i=1}^{d} w_i x_i\Big)_+, \qquad x \in \mathbb{R}_+^d, \tag{33}$$

for given strike $K > 0$ and weights $w \in \mathbb{R}_+^d$ with $\sum_i w_i = 1$. There is *no* running cost:

$$h(x, \alpha) \equiv 0.$$

The cost functional in our minimization convention is thus

$$J(\alpha) = \mathbb{E}\big[g(X_T)\big], \tag{34}$$

and we consider the problem

$$\inf_{\alpha\in\mathcal{A}} J(\alpha),$$

where $\mathcal{A}$ is the set of admissible progressively measurable $\{0, 1\}$–valued controls. Intuitively, an admissible control chooses a (possibly random) time at which to freeze the basket; the terminal payoff $g(X_T)$ is then the basket put evaluated at the frozen state.

Extended Hamiltonian with zero running cost. In the general notation of Section 2.1, we have

$$a(x, \alpha) = \alpha\, a(x), \qquad b(x, \alpha) = \alpha\, b(x), \qquad h(x, \alpha) \equiv 0.$$

Therefore the extended Hamiltonian is

$$\begin{aligned} H(x, \lambda, V, \gamma) &= \inf_{\alpha\in\{0,1\}} \Big\{\lambda^\top a(x, \alpha) + \tfrac{1}{2}\,\mathrm{Tr}\big[b(x, \alpha) b(x, \alpha)^\top V\big] + \gamma\, h(x, \alpha)\Big\} \\ &= \inf_{\alpha\in\{0,1\}} \Big\{\alpha\big(\lambda^\top a(x) + \tfrac{1}{2}\,\mathrm{Tr}\big[b(x) b(x)^\top V\big]\big)\Big\}. \end{aligned} \tag{35}$$

Since $A = \{0, 1\}$, the infimum is attained and we can write

$$H(x, \lambda, V, \gamma) = \min\big\{H_0(x, \lambda, V, \gamma),\ H_1(x, \lambda, V, \gamma)\big\}, \tag{36}$$

with

$$H_0(x, \lambda, V, \gamma) := 0 \quad (\alpha = 0), \qquad H_1(x, \lambda, V, \gamma) := \lambda^\top a(x) + \tfrac{1}{2}\operatorname{Tr}\big[b(x)b(x)^\top V\big] \quad (\alpha = 1). \tag{37}$$

In particular, with the choice $a(x) = 0_d$, $b(x) = \operatorname{diag}(x)\sigma$, we get the explicit form

$$H(x, \lambda, V, \gamma) = \min\Big\{0,\ \tfrac{1}{2}\operatorname{Tr}\big[\operatorname{diag}(x)\,\sigma\sigma^\top \operatorname{diag}(x)\,V\big]\Big\}, \tag{38}$$

which is concave in (λ, V, γ) and affine in (λ, V) on each branch.

Away from the switching manifold

$$\mathcal{S}(x) := \Big\{(\lambda, V, \gamma) :\ 0 = \lambda^\top a(x) + \tfrac{1}{2}\operatorname{Tr}\big[b(x)b(x)^\top V\big]\Big\},$$

the Hamiltonian is C^1 in (λ, V, γ) and its first derivatives are simply

$$\begin{aligned} &\text{if } H_0 < H_1 \text{ ('frozen' region)}: \quad \partial_\lambda H = 0, \quad \partial_V H = 0, \quad \partial_\gamma H = 0, \\ &\text{if } H_1 < H_0 \text{ ('active' region)}: \quad \partial_\lambda H = a(x), \quad \partial_V H = \tfrac{1}{2}b(x)b(x)^\top, \quad \partial_\gamma H = 0. \end{aligned} \tag{39}$$

The terminal payoff g enters only through the terminal condition of the HJB equation and the objective $J(\bar{u}) = \mathbb{E}[g(X_T)]$; there is no running cost contribution $\partial_\gamma H$ in this example.

Smooth soft-min regularization. Since H is the pointwise minimum of two affine branches, it is concave but non-smooth along $\mathcal{S}(x)$. We regularize it in closed form using a soft minimum based on log-sum-exp. For $\delta > 0$, define

$$H_\delta(x, \lambda, V, \gamma) := -\,\delta\,\log\Big(\exp\big(-H_0(x, \lambda, V, \gamma)/\delta\big) + \exp\big(-H_1(x, \lambda, V, \gamma)/\delta\big)\Big). \tag{40}$$

Since $H_0 \equiv 0$, this simplifies to

$$H_\delta(x, \lambda, V, \gamma) = -\,\delta\,\log\Big(1 + \exp\big(-H_1(x, \lambda, V, \gamma)/\delta\big)\Big),$$

which is C^∞ and concave in (λ, V, γ) for each fixed x, and satisfies $H_\delta \to H$ locally uniformly as $\delta \downarrow 0$.

Introduce the soft-min weights

$$w_0(x, \lambda, V, \gamma) := \frac{\exp(-H_0/\delta)}{\exp(-H_0/\delta) + \exp(-H_1/\delta)} = \frac{1}{1 + \exp(-H_1/\delta)}, \qquad w_1 := 1 - w_0 = \frac{1}{1 + \exp(H_1/\delta)}. \tag{41}$$

A standard computation yields the first derivatives of H_δ as

$$\partial_\lambda H_\delta = w_0\,\partial_\lambda H_0 + w_1\,\partial_\lambda H_1 = w_1\,a(x), \qquad \partial_V H_\delta = w_1\,\tfrac{1}{2}b(x)b(x)^\top, \qquad \partial_\gamma H_\delta = 0. \tag{42}$$

In particular, the smoothed P2 dynamics (6) driven by a potential $\bar{u}_\theta$ are

$$\mathrm{d}X_t = \partial_\lambda H_\delta\big(X_t, \nabla\bar{u}_\theta, \nabla^2\bar{u}_\theta, 1\big)\,\mathrm{d}t + \Sigma_t\,\mathrm{d}W_t, \qquad \Sigma_t\Sigma_t^\top = 2\,\partial_V H_\delta\big(X_t, \nabla\bar{u}_\theta, \nabla^2\bar{u}_\theta, 1\big),$$

that is,

$$\mathrm{d}X_t = w_1\big(X_t, \nabla\bar{u}_\theta, \nabla^2\bar{u}_\theta, 1\big)\, a(X_t)\,\mathrm{d}t + \sqrt{w_1\big(X_t, \nabla\bar{u}_\theta, \nabla^2\bar{u}_\theta, 1\big)}\, b(X_t)\,\mathrm{d}W_t.$$

With $a(x) = 0_d$, the drift vanishes and the control acts purely by modulating the diffusion intensity between full volatility ($w_1 \approx 1$) and a frozen state ($w_1 \approx 0$), while the objective remains

$$J(\bar{u}_\theta) = \mathbb{E}\big[g(X_T)\big].$$

This yields an analytically tractable, high-dimensional benchmark in which the Hamiltonian and its regularization are available in closed form and the terminal payoff is a basket American-style payoff evaluated at the frozen state.

Additional test problems: We propose three progressively more challenging test cases:

1. **Nonlinear drift + linear diffusion:** $dX_t = (AX_t + B\tanh(u_t))\,dt + \sigma\,dW_t$, with quadratic cost. This introduces a non-convex control cost and tests robustness of the minimization over H.

2. **Vanishing noise (stiff dynamics):** A problem with anisotropic diffusion $\sigma(x) \to 0$ in some directions. This tests numerical stability and sensitivity of the pathwise gradient estimates.

3. **State constraints:** Control of a process confined to a convex domain (e.g. a ball or box), with reflecting or viability conditions encoded via a constraint set K. This requires computing H_K and verifying the stability of the approximation under state constraints.

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We build on the deterministic costate/potential formulation[17] and standard references for controlled diffusions, HJB/SMP, viscosity solutions, and numerical stochastic control[4, 1, 18, 10, 11, 2, 8].

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